

Transfer Learning Between Morphologically Different Robots

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 - Teacher-Centered Transfer
 - Task-Centered Transfer
- 3 State Abstraction
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- Robotics involves complex and repetitive tasks
- Neural Networks require:
 - Large datasets
 - High computational resources (time, money, energy)
- Proposed solution: **Transfer Learning**

Transfer Learning Categories

Based on Mehdi Mounsif's thesis:

- Teacher-centered transfer
- Task-centered transfer

Strength

- Fast learning after transfer

Weakness

- Impossible to surpass the expert

Strengths

- Universal Neural Network
- Works in zero-shot scenarios
- Simple reward function (fine-tuning)

Weakness

- Restrictive interface

Transfer Learning by State Abstraction

Based on Samuel Beaussant's thesis

Strengths

- Reduced training cost
- Inter-robot generalization
- Reduced fine-tuning time

Weaknesses

- Dynamic tasks require fine-tuning
- Sensitivity to overfitting

Effect of Number of Steps

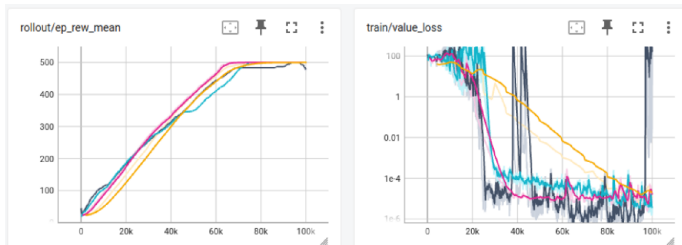
Too few steps:

- Episode mean reward : high variance and oscillation
- Lot of noise on the loss value

Many steps :

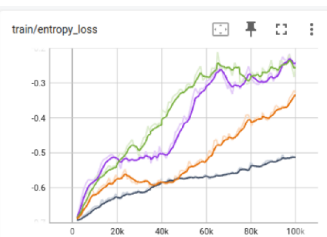
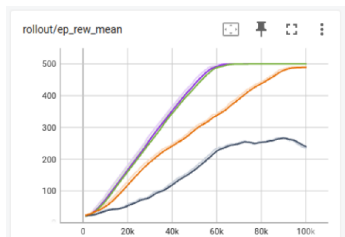
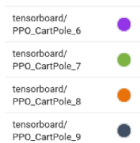
- Episode mean reward : smoother increase
- Loss value more stable

tensorboard/ PPO_CartPole_2	●
tensorboard/ PPO_CartPole_3	●
tensorboard/ PPO_CartPole_4	●
tensorboard/ PPO_CartPole_5	●



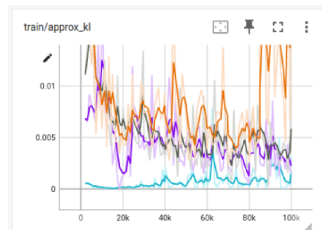
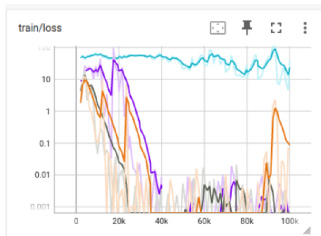
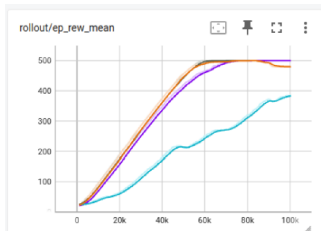
Effect of Batch Size

- Stability of gradient updates
- Small batch: faster local adaptation
- Large batch: smoother gradients



Effect of Number of Epochs

- Reuse of collected data
- Too few: underfitting
- Too many: overfitting



tensorboard/
PPO_CartPole_10

tensorboard/
PPO_CartPole_11

tensorboard/
PPO_CartPole_12

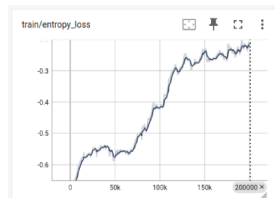
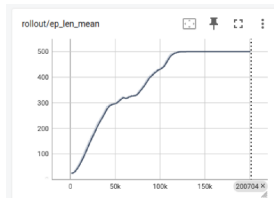
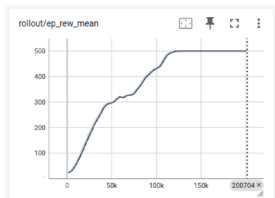
tensorboard/
PPO_CartPole_13

tensorboard/
PPO_CartPole_14

tensorboard/
PPO_CartPole_15

Effect of Total Timesteps

- Total interaction with the environment
- Too few: undertrained policy
- Too many: diminishing returns



Conclusion and Future Work

Thank you for your attention!

Any questions?